



Proposed CEA Link — December 2026

How this rota would work from December 2026, and how it compares with today's link

Prepared 29 September 2026 · every figure is calculated from the rota, not entered by hand

PROPOSAL

Targeted Fatigue Redo

A cover week at line 18, with Sunday and Monday duties placed to reduce fatigue

CODE **TF-24-EXT**

FINGERPRINT **8eef9a13**

SET AGAINST today's 20-week link

CREATED 22 September 2026

- ✓ **24** lines, one per person
- ✓ **35h 00m** a week Mon–Sat, on average — exactly the contract
- ✓ longest run **9** days (limit 13)
- ✓ **0** rests under 12h
- ⚠ **4** avoidable tiring patterns — 3 as rostered (today 5)
- ⚠ **8** of 11 December staffing rules not met

What this is. Targeted Fatigue Redo is a proposal for the CEA link on the December 2026 timetable: a 24-week rotation for 24 people, where today's link has 20. It is for discussion, not a decision — it has not yet been through the roster office or a union rep.

DOES IT MEET THE DECEMBER STAFFING LEVELS?

3 of 11

rules met — 8 not met
today's link meets 4 · each rule on page 6

CAN IT BE RUN WITHIN THE LIMITS?

Yes

inside every hard limit
shortest gap 13h35 (limit 12h) ·
most days in a row 9 (limit 13) ·
35-hour contract exact

IS IT TIRING?

4

avoidable tiring patterns
today's link has 5 · cover weeks at their worst · page 7

WHAT DOES IT TAKE?

24 people

4 more than today's 20
4 cover weeks for leave and sickness · 9 on a Sunday as overtime (today 8)

ARE STAFF WHERE THE TRAINS ARE?

Closer

than today, every day of the week
weekdays 38.5 (today 52.0) ·
Sat 20.7 (65.9) · Sun 64.8 (71.7) · 0 is a perfect match · page 5

5 in 24

full weekends off
one in 5 weeks — today one in 5 (4 in 20)

48 a year

finishes at 23:00 or later, each
today 39 · about one extra every 6 weeks

54 a year

starts at 06:20, the open, each
today 62 · about one fewer every 6 or 7 weeks

Staff are likely to welcome

- Shorter weekday closing shift
- Some Saturday closing shifts shorter
- Fewer tiring patterns — 4 (today 5)
- 13 of its 19 shift times are already worked today
- More rest between shifts — at least 13h35 (today 12h30)
- The average shift 11 minutes shorter

Staff are likely to worry about

- A tiring pattern today's link does not have: too many shifts without a two-day break
- 6 new shift times to learn
- More late finishes — about one extra every 6 weeks each
- About 5 more days at work a year

Still to settle

- **Rule not met:** At least four on at the open, every day — here 4 · 5 · 3.
- **Rule not met:** At least five still on duty at 22:00, every day — here 4–6 · 3 · 3.
- **Rule not met:** Fourteen on a Saturday, ten on a Sunday — here 12 · 9.
- **Rule not met:** Four cover weeks, evenly spread — here lines 1, 7, 12, 18.
- **Rule not met:** The ticket office: two early and two late, identical shifts, every day — here Mon–Fri no · Sat no · Sun no.
- **Rule not met:** Every weekday closer starts at 15:45 — here closers start at 16:25.
- **Rule not met:** At least two on the floor at every moment — here fewest 3 · 3 · 1.
- **Rule not met:** No more shift times than today — here 19.
- **Which link is used.** This is a proposal; colleagues' views come first.
- **Then who starts on which line.** The rota does not say who works which week.

What changes, in numbers

Today's 20-week link beside Targeted Fatigue Redo. Green is better than today, amber is worse, unshaded is neither.

QUESTION	TODAY	TARGETED FATIGUE REDO	IN PLAIN WORDS
People on the link	20	24	4 more people, one per line
On duty each day — weekday · Saturday · Sunday	11-12 · 10 · 8	13-16 · 12 · 9	the December levels ask for 14 on a Saturday and 10 on a Sunday
On at the open — weekday · Sat · Sun	4 · 4 · 3	4 · 5 · 3	at least 4 every day — not met
Still on duty at 22:00 — weekday · Sat · Sun	4 · 2 · 3	4-6 · 3 · 3	at least 5 every day — not met
Through to the close — weekday · Sat · Sun	2 · 2 · 3	3-4 · 3 · 3	at least 3 every day — met
How closely staff follow the trains — weekday · Sat · Sun	52.0 · 65.9 · 71.7	38.5 · 20.7 · 64.8	lower is closer; 0 would be a perfect match — see page 5
Cover weeks — for leave and sickness	lines 1, 7, 12, 17	lines 1, 7, 12, 18	not evenly spaced (gaps of 6, 5, 6 and 7 weeks)
Most days worked in a row	9	9	Chiltern's limit is 13 (written source to confirm)
Shortest gap between two shifts	12h30	13h35	the limit is 12 hours
Full weekends off	4 in 20	5 in 24	one in 5 weeks, against one in 5 today
Days at work a year, not counting Sundays	219	224	Sundays are overtime; a cover week counts as 4 days
Average shift · longest shift	8h14 · 9h10	8h03 · 9h10	the average is Monday to Saturday; the longest is on any day
Early shifts, shortest to longest	7h15-8h30	7h15-8h30	an early starts before 11:00 · any day of the week
Late shifts, shortest to longest	7h30-9h10	7h30-9h10	a late starts at 11:00 or after · any day of the week
Closing shift — weekday · Sat · Sun	8h40 · 9h10 · 8h55	7h30 · 7h30-9h10 · 8h55	the shift that locks up
How far the start time moves from one week to the next — on average	4h00	4h10	smaller is easier on the body clock
Finishing at 23:00 or later — each, a year	39	48	about one extra every 6 weeks
Starting at 06:20, the open — each, a year	62	54	about one fewer every 6 or 7 weeks
Saturdays worked — each, a year	26	26	a rostered Saturday is paid at time and a quarter; cover weeks not counted
Sunday overtime to share — each, a year	21	20	Sundays are overtime, as today
Different shift times	18	19 (13 worked today)	6 new to learn, on 14 of the 20 working weeks — on page 4, the times with no figure under today's link
Avoidable tiring patterns (ORR and rail-industry lists)	5	4	early starts and a weekly rotation come with every link
December staffing rules met	4 of 11	3 of 11	staffing levels confirmed verbally, 29 Sep 2026

The rest of this sheet — the workings

3 The rota — all 24 weeks, day by day, with hours and days worked

4 A week, and every shift time — beside the ones worked today

5 People on duty, hour by hour — against the December trains

6 The limits and the eleven December rules — each one, with today's link beside it

7 The fatigue checks — the Office of Rail and Road's list, in full

8 How the figures are worked out

THE ROTATION

The 24-line link

Sunday to Saturday per line; everyone moves down one line each week and line 24 goes back to line 1. Hours and days at the right, cover beneath.

LINE	SUN	MON	TUE	WED	THU	FRI	SAT	MON-SAT hours	ALL WEEK hours	DAYS WORKED Mon-Sat	TURNS Mon-Fri
1	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—	—
2	14:30 23:25	16:25 23:55	16:25 23:55	16:25 23:55	16:25 23:55	RD	RD	30h 00m	38h 55m	4	1
3	07:15 15:45	RD	08:30 16:30	06:20 14:20	06:20 14:20	06:20 14:20	06:20 14:20	40h 00m	48h 30m	5	2
4	RD	RD	06:20 14:20	08:30 16:30	RD	13:30 22:00	14:45 23:55	33h 40m	33h 40m	4	3
5	14:30 23:25	RD	14:00 22:30	14:00 22:30	14:00 22:30	16:25 23:55	RD	33h 00m	41h 55m	4	2
6	RD	06:20 14:20	07:00 15:30	07:00 15:30	07:00 15:30	08:30 17:00	RD	42h 00m	42h 00m	5	3
7	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—	—
8	07:15 15:45	06:20 13:45	RD	RD	13:30 22:00	14:00 22:30	13:30 21:00	31h 55m	40h 25m	4	3
9	RD	12:00 20:30	12:00 20:30	12:00 20:30	12:00 20:30	12:00 20:30	RD	42h 30m	42h 30m	5	1
10	RD	14:00 22:30	16:25 23:55	16:25 23:55	16:25 23:55	16:25 23:55	14:45 23:55	47h 40m	47h 40m	6	2
11	RD	07:00 15:30	07:00 15:30	RD	RD	06:20 13:35	06:20 14:50	32h 45m	32h 45m	4	2
12	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—	—
13	RD	RD	14:00 22:30	13:30 22:00	14:00 22:30	14:00 22:30	14:30 22:00	41h 30m	41h 30m	5	2
14	RD	16:25 23:55	16:25 23:55	RD	RD	07:00 15:30	06:20 14:50	32h 00m	32h 00m	4	2
15	07:15 15:45	06:20 13:45	06:20 13:35	06:20 13:35	06:20 13:35	07:00 15:30	RD	37h 40m	46h 10m	5	3
16	RD	08:30 17:00	08:30 17:00	08:30 17:00	RD	RD	RD	25h 30m	25h 30m	3	1
17	10:30 19:00	16:25 23:55	RD	RD	06:20 14:20	06:20 14:20	06:20 14:50	32h 00m	40h 30m	4	2
18	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—	—
19	14:30 23:25	RD	06:20 13:45	06:20 13:45	06:20 13:45	06:20 13:45	08:00 16:00	37h 40m	46h 35m	5	1
20	08:30 16:30	06:20 13:45	RD	RD	13:30 22:00	16:25 23:55	13:30 22:00	31h 55m	39h 55m	4	3
21	13:30 22:00	13:30 22:00	RD	13:30 22:00	13:30 22:00	13:00 21:00	RD	33h 30m	42h 00m	4	2
22	RD	14:00 22:30	13:30 22:00	13:30 22:00	13:30 22:00	RD	RD	34h 00m	34h 00m	4	2
23	RD	RD	RD	16:25 23:55	16:25 23:55	16:25 23:55	16:25 23:55	30h 00m	30h 00m	4	1
24	RD	RD	06:20 13:45	06:20 13:45	07:00 15:30	RD	06:20 13:45	30h 45m	30h 45m	4	2
On duty	9 SP 4	13 SP 4	15 SP 4	15 SP 4	16 SP 4	16 SP 4	12 SP 4	35h 00m	38h 13m	4.35	4 one-turn weeks
Minutes	4,635	6,225	7,205	7,205	7,780	7,690	5,895	42,000	46,635		

Early turns
Late turns
Cover (spare) week
— four duties of seven, any turn
Rest day

Mon-Sat hours average 35h 00m over 24 lines, a cover week counted as a contracted week · days worked average over the 20 working lines · Turns: how many different shift times the week works Monday to Friday; **bold** marks a one-turn week (see below) · Minutes: total duty minutes each day — Monday to Saturday adds up to the contract (20 people × 35 hours = 42,000 minutes)

The words on this page

Link — the whole rota: 24 weeks laid out as 24 lines. Everyone works line 1, then line 2, and so on round the wheel; the link needs 24 people.

Turn — a shift time, such as 06:20-13:35. An **early** starts before 11:00; a **late** starts at 11:00 or later.

Cover (spare) week — a week with no fixed times. You work four duties of the seven days, placed by the roster clerk to cover leave and sickness.

The contract — 35 hours a week, averaged over the 24 lines; a cover week counts as a full contracted week.

The floor — staff out on the station (gates, concourse, platforms); the ticket office joins it only at the quiet ends (page 5).

Handover — the overlap when one shift takes over from another.

Line — one week of the pattern, Sunday to Saturday. You move down a line each week, and line 24 goes back to line 1.

Rest day (RD) — a day off inside the contracted week. Two together make a proper break; one on its own does not count as a 48-hour break.

Sunday — not part of the contract. A Sunday duty is rest-day working, paid on top of the week, which is why the hours are shown Mon-Sat and all week.

One-turn week — one clock time Monday to Friday, with every day of the week an early or every day a late (shown bold in the Turns column); the easiest kind of week to live around.

Opener / closer — the first people on duty in the morning and the last at night.

Fit — how closely the number of people on duty follows the number of trains through the day; 0 is a perfect match, lower is better.

A week, and the duty table

How a working week compares with today's, and every shift time beside the ones worked today.

The shape of a week, against today

MEASURE	TODAY (20)	PROPOSED (24)
Weeks on one shift time — all earlies or all lates	7 of 16	4 of 20
Weeks mixing early and late shifts	7	6
Rest-day breaks of two days or more (of all rest-day breaks)	13 of 17	15 of 25
Single rest days between duties (not a 48h break)	4	10 (1 beside a cover week)
Days worked in a week (Sunday included; weeks × days)	1×3, 3×4, 11×5, 1×6	1×3, 6×4, 9×5, 4×6
Cover (spare) weeks (whole weeks)	lines 1, 7, 12, 17	lines 1, 7, 12, 18
Different shift times (6 not worked today)	18	19
Mon–Sat hours, average week (the contract)	35h 00m	35h 00m
Sunday duties (not contracted)	8 of 16 lines	9 of 20 lines

How many people are working

PEOPLE WORKING	TODAY	PROPOSED	WHY
Monday to Friday	11–12	13–16	24 people instead of 20, so more on each weekday
Saturday	10	12	short of the December 2026 figure of 14
Sunday	8	9	short of the December 2026 figure of 10
Ticket office Mon–Fri · Sat · Sun	4 · 4 · 3	4 · 4 · 4	rostered as pairs on Thu and Fri; the plan's posts assumed on Mon, Tue, Wed, Sat and Sun (page 6)
Opening at 06:20 (07:15 Sunday) Mon–Fri · Sat · Sun	4 · 4 · 3	4 · 5 · 3	at least four to open, every day — not met: Sunday 3
Through to the close (23:55; 23:25 Sunday) Mon–Fri · Sat · Sun	2 · 2 · 3	3–4 · 3 · 3	at least three to close, every day — met
Still on duty at 22:00 Mon–Fri · Sat · Sun	4 · 2 · 3	4–6 · 3 · 3	at least five from 22:00, every day — not met: Wed 4, Saturday 3, Sunday 3

The duty table, beside today's

SHIFT TIME	TODAY'S LINK			PROPOSED		
	MON-FRI	SAT	SUN	MON-FRI	SAT	SUN
06:20–13:35 7h 15m · Early	1		1			
06:20–13:45 7h 25m · Early	1		3 (+2)	1 (+1)		
06:20–14:00 7h 40m · Early		1		0 (-1)		
06:20–14:20 8h 00m · Early	2		2	1 (+1)		
06:20–14:50 8h 30m · Early		3		3		
07:00–15:30 8h 30m · Early			2 (+2)			
07:15–15:45 8h 30m · Early			3		3	
08:00–16:00 8h 00m · Early				1 (+1)		
08:00–16:30 8h 30m · Early	2		0 (-2)			
08:30–16:30 8h 00m · Early			1	1 (+1)		1
08:30–17:00 8h 30m · Early				1 (+1)		
10:30–19:00 8h 30m · Early						1 (+1)
11:00–19:30 8h 30m · Late	1		0 (-1)			
12:00–20:00 8h 00m · Late		1		0 (-1)		
12:00–20:30 8h 30m · Late			1 (+1)			
13:00–21:00 8h 00m · Late			1	1 (+1)		0 (-1)
13:30–21:00 7h 30m · Late		1		1		
13:30–22:00 8h 30m · Late	1		4 (+3)	1 (+1)	1 (+1)	
14:00–22:30 8h 30m · Late	2		2			
14:30–22:00 7h 30m · Late		2		1 (-1)		
14:30–23:25 8h 55m · Late			3		3	
14:45–23:55 9h 10m · Late		2		2		
15:15–23:55 8h 40m · Late	2		0 (-2)			
16:25–23:55 7h 30m · Late			4 (+4)	1 (+1)		

People on each shift time on a Saturday, a Sunday and — under Mon–Fri — the weekday with most of that time, so Mon–Fri can add up to more than work on one day. Green: more than today; amber: fewer; struck through: not used.

Monday to Friday add up to 36,105 minutes of duty and Saturday 5,895 — together exactly 42,000 minutes — 35 hours for each of the 20 working weeks, the contract (a cover week counts as a full 35-hour week on top). Sunday adds 4,635, paid on top of the contracted week. 6 of the 19 shift times are new — nobody works them today.

People on duty, hour by hour

Cover today and proposed against the measured December 2026 timetable (arrivals and departures, weighted by train length). A darker orange hour carries more of the day's trains.

How to read it: each blue square is the number of people on duty at any point in that hour (a handover inside the hour counts both people) (darker blue = more people). The orange row is the timetable — the train carriages arriving and leaving in that hour, which is how busy it is (darker orange = busier). Read the columns top to bottom: cover should be thickest where the orange is darkest. The **fit** figure at the right end of each row scores that match for the whole day.

Monday to Friday

staffed 06:20–23:55 · a row for each different weekday

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	FIT
TODAY MON · WED		4	4	6	6	6	7	7	7	7	7	7	5	5	5	4	4	4	2	54.8
TODAY TUE · THU · FRI		4	4	6	6	6	7	7	8	8	8	8	6	6	6	5	5	4	2	49.1
IN THE TICKET OFFICE		1	1	2	2	2	2	2	2	4	2	2	2	2	1	1	1	1		
ON THE FLOOR MON · WED		3	3	4	4	4	5	5	5	3	5	5	3	3	4	3	3	3	2	61.1
ON THE FLOOR TUE · THU · FRI		3	3	4	4	4	5	5	6	4	6	6	4	4	5	4	4	3	2	50.1
PROPOSED MON		4	5	6	6	6	6	7	8	7	6	8	7	7	7	7	6	5	3	28.1
PROPOSED TUE		4	6	8	8	8	8	9	10	9	8	9	7	7	7	7	6	5	3	43.4
PROPOSED WED		4	5	7	7	7	7	8	11	9	8	10	8	8	8	8	7	4	3	42.2
PROPOSED THU		4	6	6	6	6	6	7	11	11	9	10	10	10	10	10	9	5	3	48.7
PROPOSED FRI		4	6	7	7	7	7	8	10	10	8	10	9	9	9	9	7	6	4	31.3
IN THE TICKET OFFICE		1	1	2	2	2	2	2	2	4	2	2	2	2	1	1	1	1		
ON THE FLOOR MON		3	4	4	4	4	4	5	6	3	4	6	5	5	6	6	5	4	3	35.5
ON THE FLOOR TUE		3	5	6	6	6	6	7	8	5	6	7	5	5	6	6	5	4	3	44.7
ON THE FLOOR WED		3	4	5	5	5	5	6	9	5	6	8	6	6	7	7	6	3	3	46.2
ON THE FLOOR THU		3	5	4	4	4	4	5	9	7	7	8	8	8	9	9	8	4	3	66.1
ON THE FLOOR FRI		3	5	5	5	5	5	6	8	6	6	8	7	7	8	8	6	5	4	37.1
TRAIN CARRIAGES, DEC 2026	5	69	107	127	122	95	85	75	84	79	87	105	140	138	107	90	73	75	66	

Saturday

staffed 06:20–23:55

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	FIT
TODAY		4	4	4	4	4	4	5	6	9	6	6	6	6	6	5	4	2	2	43.2
IN THE TICKET OFFICE		1	1	2	2	2	2	2	2	4	2	2	2	2	1	1	1			
ON THE FLOOR		3	3	2	2	2	2	3	4	5	4	4	4	4	5	4	3	2	2	65.9
PROPOSED		5	5	6	6	6	6	6	8	10	6	6	6	6	6	6	5	3	3	18.2
IN THE TICKET OFFICE		1	1	2	2	2	2	2	2	4	2	2	2	2	1	1	1			
ON THE FLOOR		4	4	4	4	4	4	4	6	6	4	4	4	4	5	5	4	3	3	20.7
TRAIN CARRIAGES, DEC 2026	6	24	58	75	81	92	88	77	64	68	72	84	82	67	64	79	78	62	41	

Sunday

staffed 07:15–23:25

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	FIT
TODAY			3	4	4	4	4	4	5	8	8	5	4	4	4	4	3	3	3	64.5
IN THE TICKET OFFICE			1	1	2	2	2	2	2	3	3	1	1	1	1	1	1	1		
ON THE FLOOR			2	3	2	2	2	2	3	5	5	4	3	3	3	3	2	2	3	71.7
PROPOSED			3	4	4	5	5	5	6	9	9	6	5	5	4	4	4	3	3	61.6
IN THE TICKET OFFICE			1	1	2	2	2	2	4	4	4	2	2	2	2	2	2	2		
ON THE FLOOR			2	3	2	3	3	3	2	5	5	4	3	3	2	2	2	1	3	64.8
TRAIN CARRIAGES, DEC 2026			15	45	65	71	74	71	61	55	64	69	78	73	75	52	71	65	50	

Ticket office. Its staff are not floor cover, bar one of each pair at the quiet ends (until 08:00, or 09:00 Sunday; from 19:30; half on a Sunday evening). Today's office matches the plan except on **Sunday**: two earlies and **one** late (a 14:30–23:25 closer, in the office until 22:30), where the plan has two and two. This design rosters the office pairs on Thu and Fri; on Mon, Tue, Wed, Sat and Sun its floor figure is everyone on duty minus the four office posts the plan asks for. The indented rows take the office out of each side, less its quiet-end help, so the floor rows compare like with like.

Cover weeks are not in these figures — a cover week has no fixed times — so real cover is a little higher. **Fit** is one number for how closely the people on duty follow the trains through the day: 0 would be a perfect match, lower is better. Page 1's weekday figure is worked out from the five weekdays combined, so it is not an average of the scores shown. On Sundays five trains in the December 2026 timetable arrive or leave after the 23:25 finish (the last at 23:54); the bar under the Sunday 23:00 hour marks the time after the finish. No duty runs past 23:25 on a Sunday — that is agreed practice and stays so (settled 28 Sep 2026) — so those trains fall outside the staffed day of any link, today's included.

Reading it. The **floor** rows are the fair comparison. The ticket office is staffed to its opening hours, not to the trains, so counting it in can make a day look better or worse than the floor really is. The fit measures **shape, not numbers** — each hour's share of the day's staff against its share of the day's trains — so an extra person in a quiet hour makes it worse even though nobody is worse off. Read it beside the numbers of people, never alone.

Hard limits and the December 2026 rules

Hard limits first — met or not. Then the December 2026 rules, each met or not. The fatigue factors are on page 7.

Hard limits — a design either meets these or cannot be run

- ✓ **No more than 13 days worked in a row** — longest possible run **9** days (today 9)
Longest possible run is 9 days, within Chiltern's limit of 13. A cover week is 4 duties in 7 days and the roster clerk places them, so this is the worst case the link allows. Worked as a block of four rather than split day-on-day-off, the worst case is the same — **9** days. It does depend on where the roster clerk puts the four: placed well, the longest run is **8** days.
The written source of the 13-day limit is still to be confirmed.
- ✓ **At least 12 hours between duties** — **0** rests under 12h anywhere in the rotation, Saturday-into-Sunday and line-into-line included (today 0)
Tightest anywhere: **13h 35m — line 20 Fri 16:25-23:55 into Sat 13:30-22:00** (today's tightest: 12h 30m).
- ✓ **The contracted week, exactly** — **35h 00m** average Mon–Sat over the 24 lines, cover weeks counted as contracted weeks
700h of duty a week across 20 working lines. Sunday duties (77h 15m in all) are paid on top as rest-day working, as they are today. Individual weeks range 25h 30m–47h 40m; only the average is the contract.

The December 2026 rules — each one, with today's link beside it

RULE	TODAY'S LINK	THIS PROPOSAL	WHAT THE RULE ASKS
At least four on at the open, every day	× 4 · 4 · 3	× 4 · 5 · 3	from 06:20 (07:15 on a Sunday)
At least three through to the close, every day	× 2 · 2 · 3	✓ 3–4 · 3 · 3	until 23:55 (23:25 on a Sunday)
At least five still on duty at 22:00, every day	× 4 · 2 · 3	× 4–6 · 3 · 3	someone finishing at 22:00 does not count
Fourteen on a Saturday, ten on a Sunday	× 10 · 8	× 12 · 9	
Four cover weeks, evenly spread	× lines 1, 7, 12, 17	× lines 1, 7, 12, 18	evenly means every 6 lines, e.g. 1, 7, 13, 19
The ticket office: two early and two late, identical shifts, every day	× Mon–Fri yes · Sat yes · Sun no	× Mon–Fri no · Sat no · Sun no	Mon–Fri 06:20–14:20 and 14:00–22:30; Sat 06:20–14:50 and 14:30–22:00; Sun two from 07:15 and two to 22:30
Every weekday closer starts at 15:45	× closers start at 15:15	× closers start at 16:25	
At least two on the floor at every moment	✓ fewest 2 · 2 · 2	× fewest 3 · 3 · 1	the ticket office counts only while its second person helps on the floor (until 08:00 or 09:00 on a Sunday, and from 19:30); checked every five minutes
Handovers: 15 minutes to each closer, 20 in the ticket office	✓ floor met · ticket office 20 min	✓ floor met · ticket office 20 min (posts on 5 days assumed)	on a Sunday each opener also stays until 15 minutes after the last closer arrives; where the design does not roster the office pairs, the plan's posts are assumed
Sunday duties between 8h and 9h	✓ 8h 00m–8h 55m	✓ 8h 00m–8h 55m	
No more shift times than today	✓ 18	× 19	today 18

Still to settle

- **8 of the 11 December 2026 rules are not met** — At least four on at the open, every day; At least five still on duty at 22:00, every day; Fourteen on a Saturday, ten on a Sunday; Four cover weeks, evenly spread; The ticket office: two early and two late, identical shifts, every day; Every weekday closer starts at 15:45; At least two on the floor at every moment; No more shift times than today. The shaded rows above give the figures.
- **New shift times.** 6 of the 19 are times nobody works today; page 4 lists them.

ORR fatigue factors

From the ORR's good-practice guidance, *Fatigue Factors*, page 3 (December 2021). $\triangle$ present — the pattern is in this design and worth a look, not a breach · $\checkmark$ clear — it is not · $\bullet$ standing — comes with the station's hours or with any rotating link, not with this design's choices · – means it does not apply here.

These are 25 roster patterns that tend to tire people — 21 from the Office of Rail and Road's good-practice list and 4 checks from the rail industry's fatigue guidance (MRSF) — long runs of earlies, short gaps between duties, start times that jump about. For each one this table asks whether the pattern is in the rotation, today and proposed, and how big it is. The list is guidance: a factor present is a question to discuss, never a pass or a fail, and a design showing nothing is not thereby approved.

CODE	FACTOR	TODAY'S LINK	PROPOSED
FF2	Early shift starting 05:00–07:00 Time of day	$\bullet$ 24	$\bullet$ 25
FF3	Very early shift starting before 05:00 Time of day	$\checkmark$ 0	$\checkmark$ 0
FF5	Day shift over 12h Duty length	$\checkmark$ 0	$\checkmark$ 0
FF7	Early shift over 10h Duty length	$\checkmark$ 0	$\checkmark$ 0
FF8b	Fewer than 2 days rest after a block of early starts Recovery time	$\triangle$ 3	$\triangle$ 5
FF11	More than 13 consecutive shifts without a 48h break Recovery time	$\checkmark$ 12 $\checkmark$ 9 as rostered	$\triangle$ 15 $\checkmark$ 12 as rostered
FF10	More than 4 consecutive 12h day shifts Cumulative	$\checkmark$ 0	$\checkmark$ 0
FF15	More than 4 consecutive early shifts in a rotating pattern Cumulative	$\triangle$ 6	$\triangle$ 7
MRSF	More than 12 consecutive day shifts Cumulative	$\checkmark$ 9	$\checkmark$ 9
MRSF	More than 55 hours worked in any 7-day period Cumulative	$\triangle$ 60.8	$\checkmark$ 54.7
MRSF	More than 7 consecutive 8h shifts Cumulative	$\triangle$ 9	$\checkmark$ 7
FF17	Backward rotating pattern Circadian	$\checkmark$ 10 backward / 12 forward	$\checkmark$ 12 backward / 17 forward
FF18	Rotating pattern of about a week Circadian	$\bullet$ 20-line rotation · typically 4h 00m a week	$\bullet$ 24-line rotation · typically 4h 10m a week
FF19	Successive start times varying by more than 2 hours (definition to confirm) Circadian	$\triangle$ 6	$\triangle$ 8
FF13	Less than 12h rest in any 24h Intervals between duties	$\checkmark$ 0	$\checkmark$ 0

Also on the list, and applying to neither link: time of day (FF1) · duty length (FF4) · cumulative (MRSF) · night shifts (FF6, FF8, FF9, FF12, FF14, FF16, FF20).

- Avoidable tiring patterns found: 4 (today 5).
- **Standing factors ($\bullet$) are recorded for information, not counted.** FF2 comes with the station's 06:20 open, and FF18 — the ORR's concern with a rotation that changes shift type about once a week (*Managing rail staff fatigue*, 7.68) — comes with every weekly link, which moves everyone one line a week by construction; its week-to-week step is shown as information.
- **How to read the table.** The figure beside each mark is the worst case found, for today's link as for the proposal, except where a row counts only the duties the rota fixes: the 55-hour row counts a cover week as no hours, and FF8b counts none of its days as early. "As rostered" under FF11 is the worst of the places a cover week's four duties could fall when worked together, which is usual. The MRSF rows are extra checks from the rail industry's guidance on managing staff fatigue, listed beside the ORR's own. FF2 and FF15 use the ORR's early — a start from 05:00 up to 06:59, not the 11:00 used elsewhere on this sheet; the station opens at 06:20, so every link has FF2. FF11 counts shifts between two-day breaks, so a single rest day does not reset it — that is why it can be longer than the most days worked in a row.
- **Cover weeks.** A cover week's four duties can fall in different ways, and on this design that moves the most days worked in a row (8 if placed well), the run of 8-hour shifts (5 if placed well), the run of early starts, FF15 (4 if placed well), FF11 (15 if spread out, 9–12 if worked together) and the 55-hour row, which counts a cover week as no hours: 54.7–56.5 if the four are worked as 8-hour duties, which would make it present. Where they fall is for the roster office. This page is an aid to discussion, not a fatigue risk assessment.

How the figures are worked out

In plain words — what each figure means and what it does not.

Where the figures come from

Every figure is worked out from the rota itself — the 24 weeks on page 3 — by the Marylebone Roster app. None is typed in by hand, and the same calculation is run on today's link.

Cover weeks

A cover week is marked on all seven days but works four of them, placed by the roster clerk. It counts as a full contracted week of hours and as four days at work. Where a figure depends on where those four fall, the worst case is shown.

Sundays

Sunday is not in the contract: Sunday duties are overtime, as today. So "days at work" and the 35-hour week are Monday to Saturday. A Sunday off costs no annual leave.

Each person, a year

Figures given "each, a year" are averages across the whole link: every week's duties, times 52, divided by the number of people. "Days at work a year" is the days a week times $365 \div 7$. Somebody's own year depends on which week they start on.

Following the trains

For each hour, the share of the day's floor staff on duty is set against the share of the day's train movements in the December 2026 timetable. Hour by hour, the difference between the two shares is squared — so a big mismatch counts far more than several small ones — and the squares are added up and multiplied by 10,000. 0 would be a perfect match; lower is closer. On page 5 a person counts in every hour they are on duty for any part of, so a handover inside an hour counts both people. It measures the shape of the day, not a staffing requirement or passenger numbers, and it compares shares, not headcounts: an extra person in a quiet hour makes the figure worse although nobody is worse off. The ticket office is left out, except its second person helping at the quiet ends.

Rest, runs and weekends

The shortest gap is one gap — the tightest anywhere, Saturday into Sunday and the last week into the first included; a rota with many gaps just over 12 hours reads the same as one with none. The most days in a row is the worst case, with a cover week's four duties placed as badly as they can be. A full weekend off is a Saturday off followed by a Sunday off.

Familiar shift times

How many of the shift times somebody already works today. A rough guide to how much there is to learn, not to what staff will accept.

Fatigue

The Office of Rail and Road's good-practice list of roster patterns that tend to tire people, with four rail-industry checks. It is guidance, not a pass or fail, and this is not a fatigue risk assessment. Two patterns — early starts at a 06:20 station, and a weekly rotation — come with every link, today's included, so they are recorded on page 7 and not counted. In the ORR's checks an early start is one from 05:00 up to 06:59; a "block" of earlies is two or more in a row; the start-time rows count how many times a start moves by more than two hours (a count, not hours); and the week-to-week move is the average change in a week's mean start time.

What is confirmed

The December 2026 staffing levels, a 24-person link and the Sunday cover were confirmed verbally on 29 September 2026. The written source of Chiltern's 13-day limit is still to be confirmed.

How firm each figure is

Firm: the rest gaps, the contract, the days in a row and the staffing counts — worked out from the rota and checked again by the app. Confirmed verbally: the December staffing levels. Advisory: the fatigue patterns — guidance, never a pass or fail. A guide: how closely staff follow the trains, and how familiar the shift times are.

What no page can tell you

Whether staff accept the new shift times; where the roster clerk places a cover week's four duties; how leave and sickness cover land; whether the pattern holds on Wembley event days, since it follows the train timetable rather than passenger numbers; and who starts on which week. Those are for the people who work it and the managers who roster it.

The rota is supplied beside this PDF as Targeted-Fatigue-Redo-TF-24-EXT-import.txt, ready to paste into the Links page (Import), which re-runs every check here from the pasted weeks. The code **8eef9a13** in every footer is worked out from the 168 days of the rota and nothing else, so a printout always matches the design it came from.